A. Reconstruct and Challenge Natural-Language Arguments

Core vocabulary and response rule

An **argument** contains **premises** offered in support of a **conclusion**. An **intermediate conclusion** is supported by earlier claims and then supports a later conclusion. A **bridge premise** supplies a missing link. An argument is **valid** exactly when no interpretation makes all premises true and the conclusion false.

On the Core, complete all requested parts in A, but any requested explanation or justification must be only one or two sentences. Symbols, labels, and compact argument maps are encouraged.

A.1 Argument structure

Mark indicator words (*because, since, therefore, so*), then ask:

1. What is the author ultimately trying to establish?
2. Which claim is supported and then reused as support?
3. Which claims are evidence, concessions, or background?
4. What bridge would make the last support step reasonable?

An argument map such as $P_1 + P_2 \Rightarrow I$ and $I + P_3 \Rightarrow C$ records convergent or linked support without extra prose.

A.1 worked example

Encrypted backups can be restored after device loss. The lab's archives are encrypted backups. Recoverable records reduce interruption after a failure. Therefore the lab should archive every final data set.

1. Endpoint: C = archive every final data set. **2. Chain:** P_1 = encrypted backups are recoverable; P_2 = the archives are encrypted backups; therefore I = the archives are recoverable. **3. Value:** P_3 = recoverability reduces interruption. **4. Gap:** I and P_3 describe a benefit but do not yet imply a policy. Bridge: if an affordable archive protects final data and reduces interruption, the lab should use it. Thus $P_1 + P_2 \Rightarrow I$ and $I + P_3 + \text{bridge} \Rightarrow C$.

A.2 Conditionals in English

$$
\begin{aligned}
P \text{ only if } Q &: P \to Q && (Q \text{ necessary for } P),\\
P \text{ if } Q &: Q \to P && (Q \text{ sufficient for } P),\\
P \text{ iff } Q &: (P \to Q) \wedge (Q \to P),\\
P \text{ unless } Q &: \neg Q \to P \quad \text{or} \quad P \vee Q.
\end{aligned}
$$

The **contrapositive** $\neg Q \to \neg P$ is equivalent to $P \to Q$. The converse $Q \to P$ and inverse $\neg P \to \neg Q$ are not.

A.2 worked example

Let A = access is granted, T = training is complete, and S = a supervisor is present.

- "Access only if trained": $A \to T$.
- "Training and supervision are enough": $(T \wedge S) \to A$.
- "Access exactly when both hold": $A \leftrightarrow (T \wedge S)$.

Step 1: Translate each sentence before combining them. **Step 2:** from $A \leftrightarrow (T \wedge S)$, both T and S are individually necessary for A, while $T \wedge S$ is sufficient. **Step 3:** from the necessary direction $A \to T$, reverse and negate to get $\neg T \to \neg A$. Merely reversing the arrow gives $T \to A$, an unsupported converse because supervision may be absent.

A.3 Validity, countermodels, and repair

Valid rules: modus ponens ($P \to Q, P \therefore Q$), modus tollens ($P \to Q, \neg Q \therefore \neg P$), hypothetical syllogism, disjunctive syllogism, conjunction, simplification, and resolution.

For an invalid argument, give a **countermodel**: one assignment making all premises true and the conclusion false. A **minimal repair** adds or changes only what is needed to block that countermodel.

A.3 worked example

If the cache is stale, the test fails. The test failed. Therefore the cache is stale. **1. Translate:** $S \to F, F \therefore S$. **2. Compare with a rule:** modus ponens would require S, not F; this instead affirms the consequent. **3. Disprove validity:** set S = F and F = T. Both premises are true while the conclusion is false. **4. Repair:** add $F \to S$, or replace the first premise with $S \leftrightarrow F$. Either repair preserves the original conclusion S.

Frequent A errors

Do not reverse "only if." Do not call evidence the main conclusion. A countermodel must make *every* premise true and the conclusion false. A repair should close the identified gap, not merely restate the conclusion.

A readiness

I can map premises, intermediate conclusions, and the main conclusion; translate necessary and sufficient conditions; use contraposition without using a converse; and test validity with a rule or a true-premise/false-conclusion countermodel.

B. Logical Reasoning I: Anatomy and Inference

How B is graded

For every B problem, choose one answer only. **No written justification is required.** The worked explanations here teach a repeatable selection method.

B.1 Main conclusion

The **main conclusion** is the claim the author ultimately wants accepted. Ask "Why believe this?" Evidence answers that question; an intermediate conclusion is both supported and reused. A concession often begins with *although* or *admittedly*.

B.1 worked example

Weekday appointments occur when many hourly workers cannot attend. Recent evening clinics filled immediately. A nurse and funding are now available. Therefore the clinic should offer Saturday appointments.
Step 1: ask which claim the others answer *Why believe that?* **Step 2:** the timing and full evening clinics show demand; the nurse and funding remove objections. **Answer:** the Saturday recommendation is the endpoint. The other claims support it and are not themselves the main conclusion.

B.2 Role of a statement

Describe what the statement *does*: premise, intermediate conclusion, main conclusion, concession, objection, reply, example, or background. Read one sentence before and after it and identify the claim it supports or qualifies.

B.2 worked example

Digitizing the archive will interrupt access for two weeks. However, it prevents irreversible loss, and the interruption can be reduced by scanning in batches. The archive should be digitized. **Step 1:** the final sentence is the recommendation. **Step 2:** the first sentence counts against it. **Step 3:** *however* introduces the author's reply. Thus the first sentence is a **concession**: an acknowledged cost that the author answers while defending digitization.

B.3 Point at issue

Find one exact proposition P such that one speaker accepts P and the other accepts $\neg P$. Agreement about background facts is not disagreement about the inference drawn from them.

B.3 worked example

Mina: "A random sample of twelve is enough to estimate majority support." Omar: "Random selection helps, but twelve is too uncertain to determine a majority." **Step 1:** both can accept that selection was random. **Step 2:** Mina accepts and Omar rejects the sufficiency claim. They disagree whether **this sample is sufficient to determine majority support**.

B.4 Must follow

A correct answer is true in *every* situation satisfying the facts. Chain conditionals, apply contrapositives, and respect quantifiers. Reject choices that are merely possible, likely, or stronger than the premises.

B.4 worked example

Facts: every certified device was inspected; no inspected device lacks a seal; unit K is certified. **Chain:** certified $\rightarrow$ inspected and inspected $\rightarrow$ sealed, so K is inspected and then sealed. It does *not* follow that every sealed device is certified: that conclusion reverses the chain.

B.5 Parallel reasoning

Abstract the structure before comparing topics. Preserve arrow direction, negations, conjunction/disjunction, and quantifier strength.

B.5 worked example

Stimulus: $P \rightarrow Q$, $P \vee R$, $\neg Q$, therefore R. **Step 1:** modus tollens yields $\neg P$. **Step 2:** disjunctive syllogism on $P \vee R$ and $\neg P$ yields R.
Parallel: "If the pump runs, water flows. The pump runs or the valve is closed. No water flows. Therefore the valve is closed." A choice using $Q \rightarrow P$ or $\neg P \vee R$ does not match.

Useful distinctions

- **Entailed**: cannot be false when the premises are true.
- **Supported**: made more credible, but not guaranteed.
- **Consistent**: can be true together; neither must imply the other.
- **Scope**: words such as *all, some, most, only, unless* control how strong a claim is.

Frequent B errors

Do not choose a vivid detail instead of the conclusion. For point-at-issue questions, both speakers must take opposite positions on the exact claim. For must-follow questions, do not add ordinary-world assumptions. For parallel reasoning, match form rather than subject matter.

B readiness

I can distinguish conclusion, evidence, concession, and role; isolate a precise disagreement; identify what must follow; and match an argument's logical form. On the Core I record only the selected answer.

C. Logical Reasoning II: Assumptions and Evidence

How C is graded

For every C problem, choose one answer only. **No written justification is required.** Use the tests below privately to select the answer.

C.1 Necessary assumption

A necessary assumption is a condition the argument *requires*. Use the **negation test**: negate a candidate precisely. If the reasoning collapses, the candidate was necessary. The assumption need not prove the conclusion.

C.1 worked example

Argument: encrypted backups can restore any file in the latest completed backup; therefore every current project file could be restored tonight. **Gap:** the premise covers backed-up files, while the conclusion covers every current file. Candidate assumption: **every current file is included in that completed backup**. **Negation test:** some current file is absent. The stated restore ability then does not cover that file, so this reasoning no longer establishes the conclusion; another recovery method might still exist.

C.2 Sufficient assumption

A sufficient assumption, when added, makes the conclusion follow. Test by inserting the choice into the premises and completing a valid chain. It may be stronger than necessary, but it must not leave a gap.

C.2 worked example

Facts: Plan K is the only plan meeting both budget and deadline. Conclusion: choose K. **Gap:** feasibility alone does not imply that a choice must be made. Add: **whenever at least one plan meets both constraints, choose a plan that meets both**. K meets them and no other plan does, so this principle requires choosing K. A rule saying only that acceptable plans must meet the constraints would not establish that K is acceptable.

C.3 Diagnose the flaw

State the gap specifically. Common flaws include converse error; confusing correlation with causation; unrepresentative sample; circular reasoning; part-to- whole or whole-to-part error; false dilemma; equivocation; and treating lack of evidence as evidence of absence.

C.3 worked example

The argument observes that benches were installed and attendance later rose, then concludes that the benches caused the increase. **Step 1:** the evidence establishes sequence and association. **Step 2:** causation would require ruling out other changes or comparing with a control. The flaw is inferring causation from an uncontrolled before–after pattern.

C.4 Strengthen

Find the conclusion and its weakest support link. A strengthener may rule out an alternative explanation, validate a measurement, make a sample representative, establish a missing connection, or show the proposed mechanism operates. It need not prove the conclusion.

C.4 worked example

Claim: high-contrast labels reduce sorting errors. Evidence: one office improved after adopting them. **Weak link:** some other change may explain the drop. **Best strengthener:** comparable lines were randomly assigned old or high-contrast labels, and only the latter showed a sustained error drop. The control and random assignment directly support the proposed causal link.

C.5 Weaken

A weakener damages the inference, not necessarily the conclusion. Look for a confounder, selection effect, bad proxy, reversed causation, missing necessary step, or evidence that the mechanism will not operate.

C.5 worked example

Claim: forcing applicants to open an accurate tutorial ensures correct forms. **Proposed mechanism:** opening → reading and understanding → correct completion. **Best weakener:** applicants can close the tutorial immediately without reading it. That breaks the first link; accuracy alone does not ensure exposure, understanding, or use.

Necessary versus sufficient

A necessary assumption N is required by the *argument's support link*: negating N undermines that reasoning. This need not make the conclusion false; it might be true for a different reason. A sufficient assumption S supplies enough for C: premises plus S imply C. Do not confuse a condition needed by this argument with a condition needed for the conclusion to be true at all.

Frequent C errors

The negation of "all" is "at least one not," not "none." A necessary assumption can be modest; a sufficient assumption can be strong. Strengtheners and weakeners target the support link, not whether you personally agree with the conclusion. Name the passage-specific flaw, not only a memorized label.

C readiness

I can apply the negation test; distinguish necessary from sufficient assumptions; name a precise inferential gap; and choose evidence that changes the actual support link. On the Core I record only the selected answer.

D. Principles and Explanations

How D and E are graded

In D, choose one answer only; **no written justification is required**. In E, complete every requested part, but keep any requested explanation or justification to **one or two sentences**. Concise labels and bullets count.

D.1 Principle support

Translate each principle as conditions → judgment. Match *every* operative condition to a stated fact; reject rules that are too broad, too narrow, or point toward the opposite judgment.

D.1 example

A fee may be waived when a person took every reasonable step but an agency outage prevented timely filing. **Conditions:** reasonable effort and an external outage. **Case check:** logs establish both for Mira. **Result:** the matching principle permits a waiver because reasonable compliance was blocked by circumstances outside the applicant's control.

D.2 Resolve the discrepancy

Treat both observations as true. Find a third fact that separates the measures, groups, or time periods so the apparent conflict disappears.

D.2 example

Fewer people registered, yet more seats were occupied. **Step 1:** keep both facts. **Step 2:** distinguish registration from attendance. **Resolution:** the new group had a much lower no-show rate, so a smaller registered group can still produce more occupied seats.

D.3 Parallel flaw

Abstract the invalid move. Example: "certified → inspected; inspected; therefore certified" affirms the consequent. The correct answer must repeat $P \to Q, Q \therefore P$, not merely reach a dubious conclusion.

D.4 Method of reasoning

Describe the argument's actual operation: applies a rule, eliminates alternatives, uses an analogy, gives a counterexample, distinguishes cases, concedes then responds, or shows that a proposal defeats its own goal.

D.4 example

An author rejects a claim that deadlines are never fair by presenting a deadline needed to coordinate emergency staffing. **Pattern:** the original claim is universal, while one fair deadline makes it false. Method: **counterexample to a universal claim**, not proof that every deadline is fair.

D.5 Complete the argument

Insert a claim that connects established facts to the stated conclusion. Prefer the weakest answer that closes the exact gap; avoid a new topic or an unnecessary universal rule.

D.5 example

Facts: a sample contains only early riders, while later riders face different service. Conclusion: the survey cannot justify system-wide cuts. **Gap:** the evidence represents only one systematically different subgroup. **Completion:** a sample limited to such a subgroup cannot support a system-wide estimate. That sentence connects the sample fact directly to the scope of the conclusion.

E. Evaluation and Revision

E.1 Causal evaluation

Separate **association** from **causation**. Name the conclusion, evidence, and a plausible confounder, selection effect, reverse cause, or measurement problem. Strong evidence uses random assignment, a valid comparison group, baseline measures, or converging mechanisms.

E.1 compact full-credit model

Claim: reminders cause timely submission; evidence: students who enabled them submitted more often. **Step 1:** this is observational and self-selected. **Step 2:** already-organized students may both enable reminders and submit on time. **Step 3:** randomly assign comparable students to reminders or no reminders and compare submission rates. The original evidence supports an association, not yet the causal claim.

E.2 Apply a principle

List the rule's conditions, compare each with the case, then state the result. Do not silently add an exception. If a term is ambiguous, state one reasonable interpretation in a short sentence.

E.2 compact full-credit model

Principle: a penalty based on a high-false-positive method requires reliable independent corroboration. **Conditions:** proposed penalty; unreliable screen; independent corroboration required. **Case check:** only the screen is positive, so corroboration is absent. **Result:** the penalty is not justified, although a proportionate investigation may still be justified.

E.3 Integrated Op-Ed analysis and revision

State the main conclusion; map the strongest support; identify the most important gap; then revise the conclusion to the strength the evidence warrants. A good revision preserves useful evidence while adding scope, comparison, or uncertainty.

E.3 compact full-credit model

Evidence: one bus-lane pilot reduced corridor travel time and coincided with higher ridership. Claim: convert every city road. **Step 1:** the evidence comes from one corridor and does not rule out other causes of ridership. **Step 2:** the conclusion jumps from one pilot to every road. **Revision:** test comparable corridors and measure travel, safety, and spillovers before deciding whether broader conversion is warranted.

Frequent D–E errors

Do not choose a principle missing one stated condition. Resolve a discrepancy without denying either observation. Match a flaw's form, not its topic. In E, do not write an essay: identify the decisive gap and the evidence or revision that addresses it in one or two sentences.

Final readiness check

- A: I can reconstruct, translate, test, countermodel, and repair.
- B–D: I can select the answer using the named test; I know no written justification is required.
- E: I can evaluate causation, apply a principle, or revise an argument in a complete but compact one- or two-sentence response.
- I can distinguish what is entailed, merely supported, and merely possible.